



TERMS OF REFERENCE (TOR)

NikoPay – Making Stablecoins Spendable

1. Background

NikoPay is a fintech startup focused on bridging stablecoins and local payment networks in Africa, beginning with Rwanda. As digital assets continue to gain adoption among freelancers, remote workers, tourists, investors, and cross-border communities, there remains a significant gap between holding digital assets and using them in everyday life.

While several platforms enable users to buy, sell, and store cryptocurrencies, very few solutions allow users to seamlessly spend stablecoins within local economies. Individuals receiving income in stablecoins often face challenges converting their digital assets into local currency for daily expenses such as rent, food, transportation, utilities, and peer-to-peer transfers.

NikoPay seeks to address this challenge by providing a secure and compliant infrastructure that enables users holding stablecoins to instantly convert them into local currency through existing mobile money networks.

The initial launch market will be Rwanda, leveraging the country's strong mobile money ecosystem and growing digital economy.

2. Vision

To make digital assets spendable across Africa by connecting stablecoins with everyday payment networks.

3. Mission

To provide a seamless bridge between blockchain-based value and local financial systems, enabling anyone holding stablecoins to access real-world purchasing power instantly and securely.

4. Project Overview

NikoPay will initially operate as a Stablecoin-to-Mobile-Money payment platform.

The platform will enable users holding supported stablecoins in external wallets such as Trust Wallet, MetaMask, Coinbase Wallet, and other WalletConnect-compatible wallets to transfer stablecoins to NikoPay and facilitate payouts directly to Mobile Money recipients in local currency.

The platform will not initially function as a custodial exchange, crypto trading platform, or wallet provider. Users will retain custody of their assets until the moment a transaction is initiated.

The Minimum Viable Product (MVP) will focus exclusively on the following transaction flow:

User Wallet → NikoPay → Mobile Money Recipient

5. Project Objectives

The primary objective of this project is to design and develop a secure digital platform that enables stablecoin holders to convert their assets into local currency and distribute funds through Mobile Money networks.

Specific objectives include:

- Enabling stablecoin transfers from external wallets.
- Facilitating Mobile Money payouts to recipients in Rwanda.
- Providing transparent exchange rates and transaction fees.
- Maintaining a treasury management system for liquidity operations.
- Providing real-time transaction monitoring and status tracking.
- Establishing an administrative dashboard for operational management.
- Creating a scalable foundation for future expansion into merchant payments, stablecoin on-ramps, and cross-border settlements.

6. Scope of Work

The development team shall design, develop, test, and deploy the NikoPay MVP platform.

The solution shall include:

User Portal

The user-facing platform shall provide the ability to:

- Connect supported cryptocurrency wallets.
- Initiate stablecoin payment requests.
- Enter recipient Mobile Money numbers.
- View exchange rates before transaction confirmation.
- Track transaction status.
- Access transaction history.

- Download transaction receipts.

Treasury Management System

The platform shall provide mechanisms for:

- Receiving stablecoin payments into designated treasury wallets.
- Monitoring wallet balances.
- Monitoring Mobile Money liquidity balances.
- Calculating settlement amounts.
- Managing payout workflows.
- Maintaining transaction records and audit trails.

Administrative Portal

The administration dashboard shall provide:

- User management.
- Transaction monitoring.
- Exchange rate management.
- Treasury balance monitoring.
- Liquidity management.
- Manual transaction review and approval capabilities.
- Risk and fraud monitoring tools.
- Reporting and analytics.

7. Functional Requirements

The system shall support the following capabilities:

Wallet Connectivity

The platform shall support external wallet integrations through WalletConnect.

Initial support shall focus on:

- USDT (TRC20)

Future support may include:

- USDC
- Polygon-based stablecoins
- Bitcoin Lightning
- Additional blockchain networks

Transaction Processing

The platform shall:

- Generate unique transaction references.
- Detect incoming stablecoin transfers.
- Validate transaction confirmations.
- Calculate payout amounts.
- Execute Mobile Money settlements.
- Notify users upon transaction completion.

Exchange Rate Engine

The system shall:

- Manage configurable exchange rates.
- Apply transaction spreads.
- Apply service fees.
- Display net payout values before transaction confirmation.

Notifications

The platform shall provide:

- Transaction status updates.
- Successful payout confirmations.
- Failed transaction notifications.
- Administrative alerts.

8. Non-Functional Requirements

The platform shall be designed with the following principles:

Security

- Secure wallet management.
- Encrypted data storage.
- Role-based access control.
- Audit logging.
- Secure API communication.
- Fraud detection mechanisms.

Performance

- High availability.
- Fast transaction processing.
- Scalable architecture.
- Efficient transaction monitoring.

Reliability

- Automated recovery mechanisms.
- Transaction reconciliation.
- Monitoring and alerting systems.
- Data backup and disaster recovery procedures.

9. Proposed System Architecture

Frontend:

- Next.js
- TypeScript
- Tailwind CSS

Backend:

- Node.js
- Express.js or NestJS

Database:

- PostgreSQL

Blockchain Services:

- TRON Network Integration
- WalletConnect

Payment Infrastructure:

- MTN Mobile Money Integration
- Airtel Money Integration (Future)

Cloud Infrastructure:

- AWS
- DigitalOcean
- Azure

10. Treasury Model

The platform shall operate using a treasury-based settlement model.

NikoPay shall maintain:

- Stablecoin Treasury Wallets

- Mobile Money Liquidity Accounts
- Settlement Records
- Internal Transaction Ledger

Customer stablecoin payments shall be received into treasury wallets, while local currency payouts shall be executed using available Mobile Money liquidity.

Treasury balances shall be monitored continuously to ensure adequate liquidity for settlement operations.

11. Deliverables

The development team shall provide:

- Functional web application.
- Administrative dashboard.
- Wallet integration module.
- Transaction processing engine.
- Mobile Money payout integration.
- Treasury management module.
- Technical documentation.
- Deployment documentation.
- Source code repository.
- Testing and quality assurance reports.

12. Future Expansion Roadmap

Following successful MVP deployment, NikoPay intends to expand into:

- Mobile Money to Stablecoin services.
- Merchant payment solutions.
- QR code payments.
- Cross-border remittances.
- Business payment APIs.
- Stablecoin payroll services.
- Regional expansion across East Africa.

13. Expected Outcome

The successful implementation of NikoPay will establish a secure and scalable payment infrastructure that transforms stablecoins into practical spending instruments, enabling users to convert digital assets into real-world purchasing power through local payment networks.

By making stablecoins spendable, NikoPay aims to become a foundational layer connecting the digital asset economy with everyday financial activity across Africa.

